Imperial College London

Introduction to 3rd year laboratory, 2025-2026.

James McGinty
Head of 3rd Year Lab.
james.mcginty@imperial.ac.uk

Mark Neil, Micros Course
mark.neil@imperial.ac.uk

Imperial College London

Today's presentation …….

- Where?
- Who?
- When?
- Safety
- Lab books
- Ethos and expectation

Imperial College
London

Experiment Locations

Experiment	Room No.
A1 – Photoelectric Effect	410A
A2 – Radiation Laws	403
B1 – Waves on a String	403
B2 – Astro Image Processing	403
C1 – Hall Effect	403
C2 – Micromagnetics	403
D1 – LEDs and Diodes	403
D2 – Laser Spectroscopy	410B
E1 – Compton Effect	409
E2 – X-ray Diffraction	403
F1 – Solar Radiation	403
F2 – Wind Turbulence	403
G – Microprocessors	418

Imperial College London

Heads of Experiments

A - Gavin Davies

B – James McGinty

C – Michael McCann

D – Stefan Truppe

E – Alex Tapper

F – Mark Richards

G - Mark Neil

Imperial College London

Demonstrators and Contact Details

		Head of Experiment	Demonstrator	Demonstrator
A	The Photoelectric Effect Radiation Laws	Gavin Davies	Cesar Silva de Freitas	Daleep Bahra
B	Wave on a String Astro Image Processing	James McGinty	Sean Lim	Yvonne Unruh
C	Matter-Antimatter Asymmetry Measuring Neutrino Oscillations	Michael McCann	Alex Hergenhan	Boyuan Shi
D	Characterising LEDs and Diodes Laser Spetroscopy	Stefan Truppe	Dimitrie Cielecki	Toby Farchy Joseph Cox
E	The Compton Effect X-ray Diffraction	Alex Tapper	Per Jonsson	Kai Hong Law
F	Solar Radiation Wind Turbulence	Mark Richards	Howard Chen	Michael Fox
G	Microprocessors	Mark Neil	Paul Dauncey	Ricardo Jose Mota Peres Tingjun Zheng

Name	Email
Daleep Bahra	d.bahra24@imperial.ac.uk
Howard Chen	yihao.chen20@imperial.ac.uk
Dimitrie Cielecki	dimitrie-calin.cielecki17@imperial.ac.uk
Joseph Cox	j.cox24@imperial.ac.uk
Paul Dauncey	p.dauncey@imperial.ac.uk
Gavin Davies	g.j.davies@imperial.ac.uk
Tobias Farchy	tobias.farchy19@imperial.ac.uk
Michael Fox	michael.fox@imperial.ac.uk
Alex Hergenhan	a.hergenhan24@imperial.ac.uk
Per Jonsson	per.jonsson@imperial.ac.uk
Kai Hong Law	kai.law16@imperial.ac.uk
Sean Lim	sean.lim21@imperial.ac.uk
Michael McCann	m.mccann@imperial.ac.uk
James McGinty	james.mcginty@imperial.ac.uk
Ricardo Jose Mota Peres	r.motaperes@imperial.ac.uk
Mark Neil	mark.neil@imperial.ac.uk
Mark Richards	mark.richards@imperial.ac.uk
Boyuan Shi	boyuan.shi18@imperial.ac.uk
Cesar Silva de Freitas	c.freitas24@imperial.ac.uk
Alex Tapper	a.tapper@imperial.ac.uk
Stefan Truppe	s.truppe@imperial.ac.uk
Yvonne Unruh	y.unruh@imperial.ac.uk
Tingjun Zheng	tingjun.zheng18@imperial.ac.uk

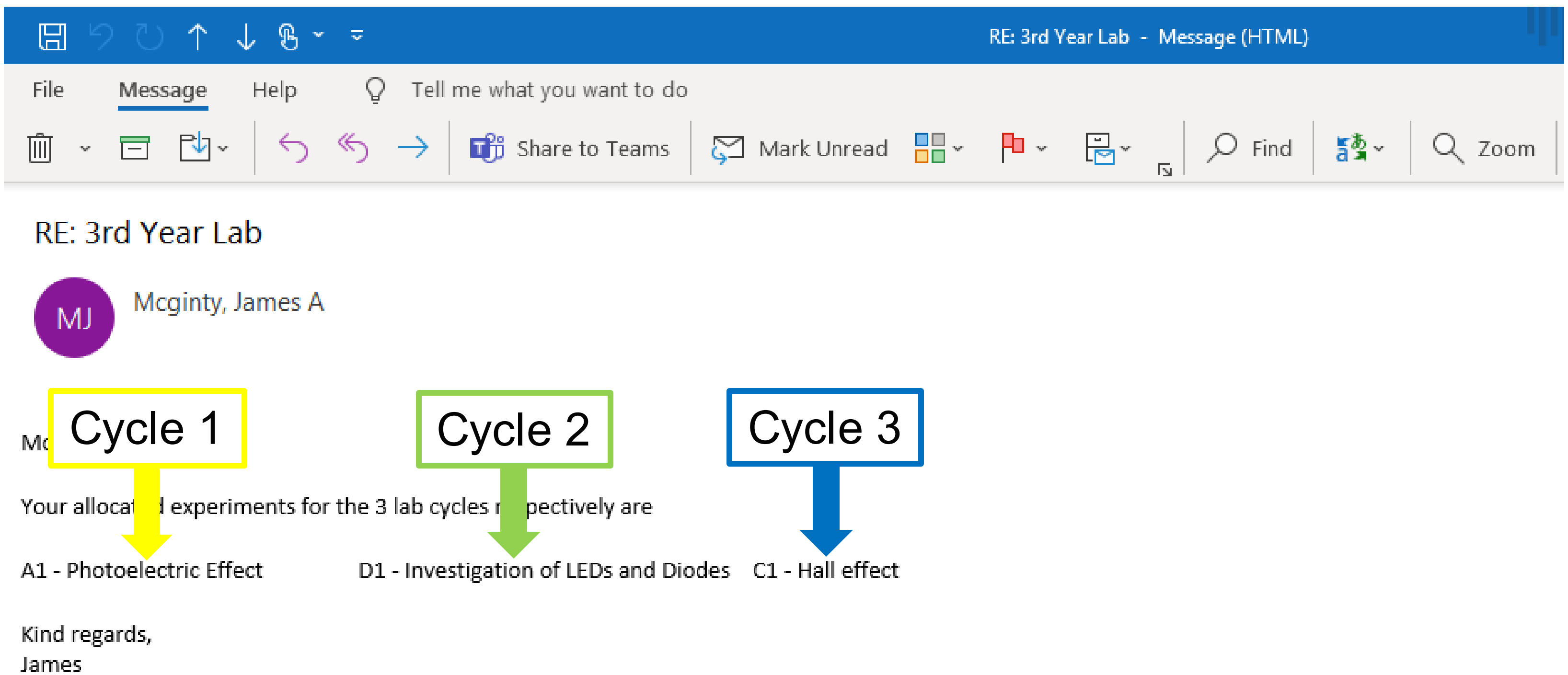
Available on Blackboard

Lab Technicians:

Graham Axtell (Head Technician)
Lee Parker
Jayesh Hirani

They are not demonstrators!

Experiment Allocation



Imperial College London

During a lab cycle

Lab sessions

- 09:00-12:00 in Tuesdays, Thursdays and Fridays
- Sign-in at the start of every session
  - tap swipecard on reader

- Let Head of Experiment know of planned absence
- Lab is open during normal working hours (~09:00-17:00)

Imperial College London

Current Cycle and Assessment Structure

LAB TIMETABLE 23/24

YEAR 3/4 TERM 1

Week 1	Time	Mon	Tues	Wed	Thur	Fri
02-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 2	Time	Mon	Tues	Wed	Thur	Fri
09-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 3	Time	Mon	Tues	Wed	Thur	Fri
16-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

- Intro Lecture
- Cycle 1
- Cycle 2
- Cycle 3
- First/last cycle session
- Online debrief
- Prepare presentation
- Presentation assessment
- 2 page submission
- 6 page submission

Week4	Time	Mon	Tues	Wed	Thur	Fri
23-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 5	Time	Mon	Tues	Wed	Thur	Fri
30-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 6	Time	Mon	Tues	Wed	Thur	Fri
06-Nov	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 7	Time	Mon	Tues	Wed	Thur	Fri
13-Nov	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 8	Time	Mon	Tues	Wed	Thur	Fri
20-Nov	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

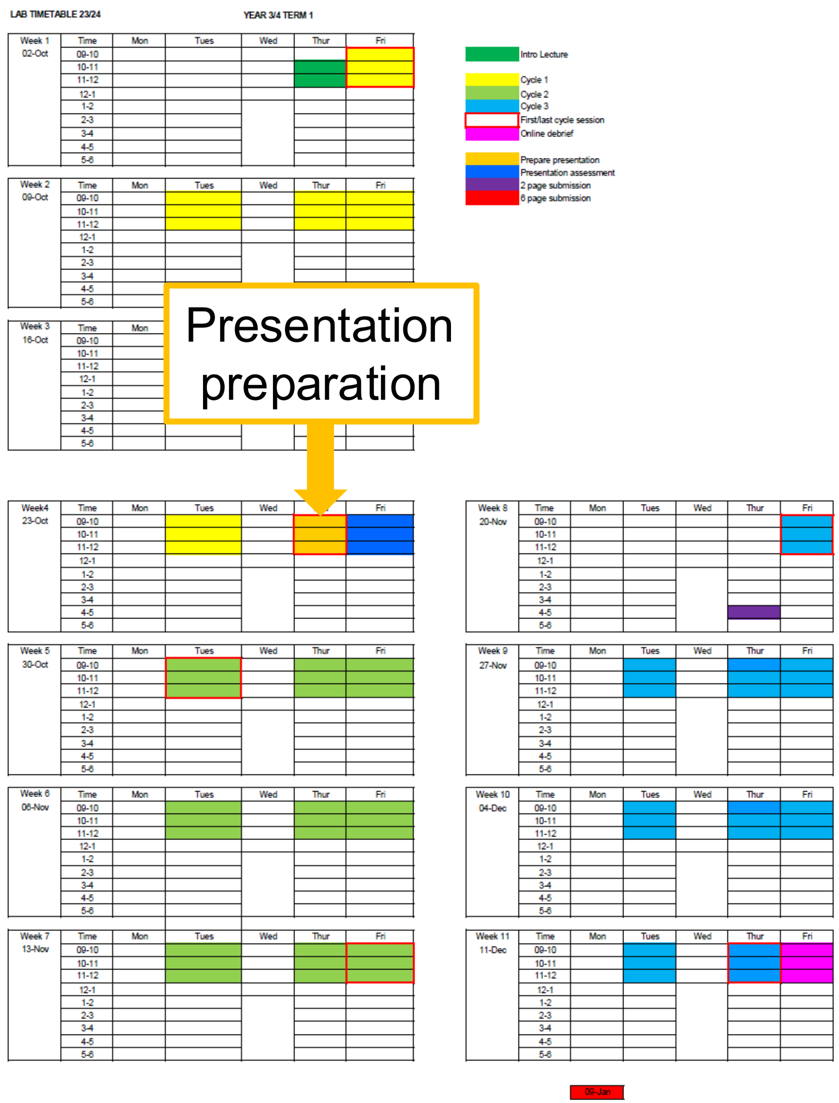
Week 9	Time	Mon	Tues	Wed	Thur	Fri
27-Nov	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 10	Time	Mon	Tues	Wed	Thur	Fri
04-Dec	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 11	Time	Mon	Tues	Wed	Thur	Fri
11-Dec	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

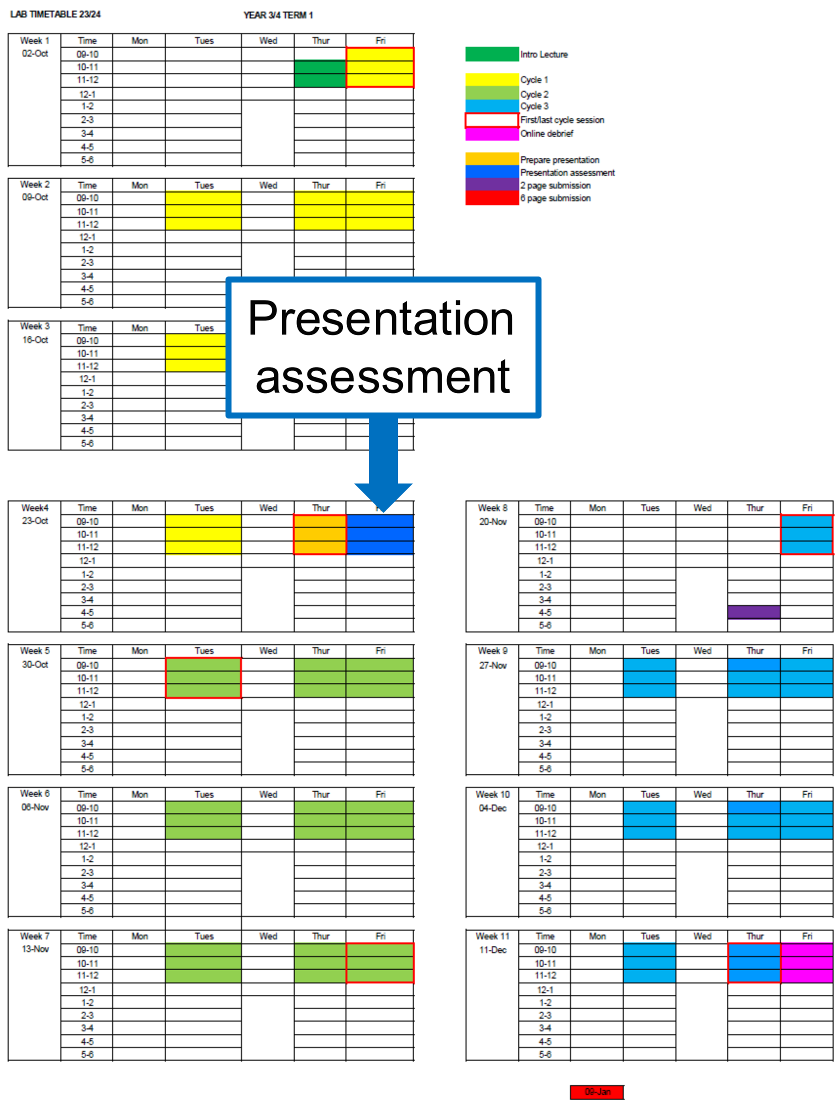
09-Jan

Current Cycle and Assessment Structure – cycle 1



Imperial College
London

Current Cycle and Assessment Structure – cycle 1



Current Cycle and Assessment Structure – cycle 2

LAB TIMETABLE 23/24

YEAR 3/4 TERM 1

Week 1	Time	Mon	Tues	Wed	Thur	Fri
02-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 2	Time	Mon	Tues	Wed	Thur	Fri
09-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 3	Time	Mon	Tues	Wed	Thur	Fri
16-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

- Intro Lecture
- Cycle 1
- Cycle 2
- Cycle 3
- First/last cycle session
- Online debrief
- Prepare presentation
- Presentation assessment
- 2 page submission
- 6 page submission

Week4	Time	Mon	Tues	Wed	Thur	Fri
23-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 5	Time	Mon	Tues	Wed	Thur	Fri
30-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 6	Time	Mon	Tues	Wed	Thur	Fri
06-Nov	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 7	Time	Mon	Tues	Wed	Thur	Fri
13-Nov	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 8	Time	Mon	Tues	Wed	Thur	Fri
20-Nov	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 9	Time	Mon	Tues	Wed	Thur	Fri
27-Nov	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Final lab session

k 10	Time	Mon	Tues	Wed	Thur	Fri
Dec	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 11	Time	Mon	Tues	Wed	Thur	Fri
11-Dec	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

09-Jan

Imperial College London

Current Cycle and Assessment Structure – cycle 2

LAB TIMETABLE 23/24

YEAR 3/4 TERM 1

- Intro Lecture
- Cycle 1
- Cycle 2
- Cycle 3
- First/last cycle session
- Online debrief
- Prepare presentation
- Presentation assessment
- 2 page submission
- 6 page submission

Week 1 02-Oct	Time	Mon	Tues	Wed	Thur	Fri
	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 2 09-Oct	Time	Mon	Tues	Wed	Thur	Fri
	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 3 16-Oct	Time	Mon	Tues	Wed	Thur	Fri
	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week4 23-Oct	Time	Mon	Tues	Wed	Thur	Fri
	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 5 30-Oct	Time	Mon	Tues	Wed	Thur	Fri
	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 6 06-Nov	Time	Mon	Tues	Wed	Thur	Fri
	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 7 13-Nov	Time	Mon	Tues	Wed	Thur	Fri
	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Submit 2 page report

Week 8 20-Nov	Time					
	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

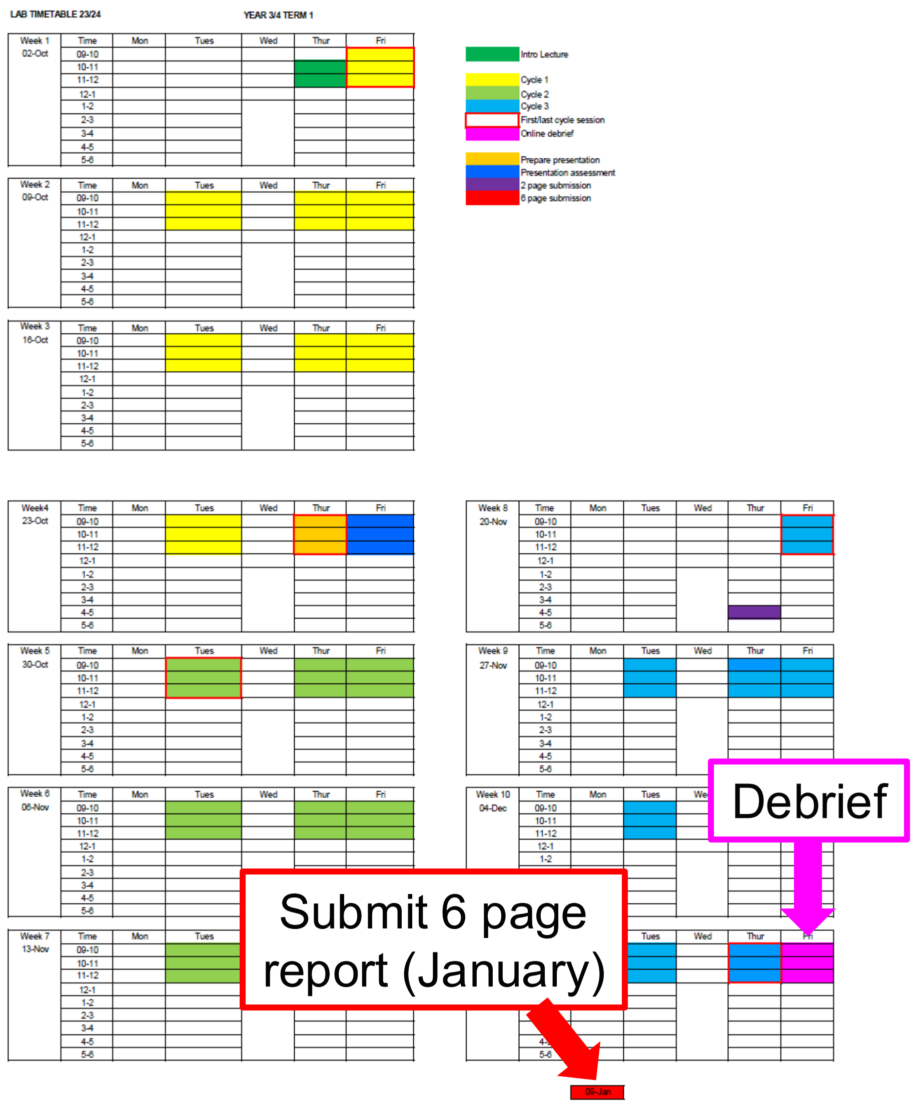
Week 9 27-Nov	Time	Mon	Tues	Wed	Thur	Fri
	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 10 04-Dec	Time	Mon	Tues	Wed	Thur	Fri
	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 11 11-Dec	Time	Mon	Tues	Wed	Thur	Fri
	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

09-Jan

Current Cycle and Assessment Structure – cycle 3



Current Cycle and Assessment Structure – Microprocessors

LAB TIMETABLE 23/24

YEAR 3/4 TERM 1

Week 1	Time	Mon	Tues	Wed	Thur	Fri
02-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 2	Time	Mon	Tues	Wed	Thur	Fri
09-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 3	Time	Mon	Tues	Wed	Thur	Fri
16-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

- Intro Lecture
- Cycle 1
- Microprocessors
- First/last cycle session
- Online debrief
- Prepare presentation
- Presentation assessment
- 2 page submission
- 6 page submission

Week4	Time	Mon	Tues	Wed	Thur	Fri
23-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 5	Time	Mon	Tues	Wed	Thur	Fri
30-Oct	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 6	Time	Mon	Tues	Wed	Thur	Fri
06-Nov	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 7	Time	Mon	Tues	Wed	Thur	Fri
13-Nov	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 8	Time	Mon	Tues	Wed	Thur	Fri
20-Nov	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 9	Time	Mon	Tues	Wed	Thur	Fri
27-Nov	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 10	Time	Mon	Tues	Wed	Thur	Fri
04-Dec	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

Week 11	Time	Mon	Tues	Wed	Thur	Fri
11-Dec	09-10					
	10-11					
	11-12					
	12-1					
	1-2					
	2-3					
	3-4					
	4-5					
	5-6					

09-Jan

Imperial College London

Safety in the laboratory

- Each Y3 Lab experiment has a Risk Assessment analysis form (a legal requirement in a UK workplace)
- This details the risks (high voltage, high temperature, UV light source, X-rays, cryogen, etc) associated with that experiment
- Before starting the experiment you must:
    - read and discuss the contents of the Risk Assessment form with your Demonstrator during first lab session

Safety in the laboratory

- **<u>NO</u> eating or drinking. Drink and foodstuffs not allowed in Lab (go out to take a coffee break).**
- Never work alone, work with Lab partner or ensure that a demonstrator or technician is on site.
- Inform the Technician(s) of your presence.
- Emergency (*medical only 09:00-17:00) - ext 4444
- Use specific safety equipment provided, e.g. gloves for liquid nitrogen.
- **All bags, coats, stools, etc, to be stored safely (do not block walkways) – ideally leave in your locker**

Cycle 1 Assessment - Presentation

- 10 minute joint presentation
- 3-5 minutes discussion/questions
- Both partners must contribute ~equally to presentation
- Must include more advanced aspects of experiment
- Given on Teams
- Send a copy of the talk to the Head of Experiment (afterwards)
- Assessed together (i.e. you get the same mark)
- 25% of lab module mark

Imperial College London

Cycle 2 Assessment – 2 page report

- 2 page conference style abstract
- Written independently
- Follow an IEEE template (on Blackboard)
- Assessed individually
- 25% of lab module mark

Cycle 2 Assessment – 6 page report

- 6 page journal style article/paper
- Written independently
- Follow an IEEE template (on Blackboard)
- Assessed individually
- 50% of lab module mark

Assessment Criteria

- Marking criteria
  - Aims
    - Structure
    - Independent thought
  - Overall Communication
  - Methods
    - Independent thought
    - Justification
  - Data
  - Conclusions
    - Relevance / insight
    - Justification

Rubric, marking and feedback

Marking Rubric – Year 3 Lab Reports – 2024 v.0.4

Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		1st			
Aims											
		Partial	Complete	Partial	Complete	Partial	Partial	Partial		Complete	
1. Structure	It is not articulated what the work was aiming to achieve.	Vague aims are provided such that success in achieving those aims is open to interpretation.		Aims are provided that specify a threshold for success.		Aims are broken down into **specific, answerable** research question(s).		All the research questions are articulated in a way which clearly allows them to be answered **quantitatively**.			
Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		Lower 1st		Upper 1st	
		Partial	Complete	Partial	Complete	Partial	Partial	Partial	Complete	Partial	Complete
2. Independent Thought (Aims)	There was no aim or research question provided.	The research question was outside the scope of physics accessible with the equipment available.		The research question was based entirely on a task given in the lab script.		The research question was an incremental variation from the lab script (e.g., probing an assumption in the lab script).		The research question was about (sensible) physics that is rarely considered with this equipment.		The research question was about (sensible) physics that has not been studied before with this equipment and is very different from the suggested directions in the lab script.	
Overall Communication											
Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		1st			
		Partial	Complete	Partial	Complete	Partial	Complete	Partial		Complete	
3. Concise transmission of information (includes evaluation of the abstract)	Many statements are made that are unnecessarily repeated and/or are irrelevant to the research question.	The structure of the account means it is difficult to understand what was done. E.g., sentences contain relevant content but more than one idea; sections/paragraphs present disconnected information.		The account provides minimal information to understand what was done such that in places there remains some ambiguity. Statements are made that are missing citations.		Sufficient information, figures, and citations are provided to clearly understand what was done, such that the reader could **reproduce the work**.		**Only** information and figures directly relevant to answering the research question(s) using the chosen method are provided. Other information is given by relevant citations to **academic** sources. ----- It is not possible to meet this category if the page/time limit is exceeded.			
Methods											
Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		Lower 1st		Upper 1st	
		Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete
4. Independent Thought (Methods)	The methods were not described or were incomprehensible.	The methods and protocols used do not go beyond details provided in the lab script.		The methods follow closely to approaches described in the lab script and provide some additional detail on intermediate steps.		Required for Upper 2^{nd} and higher: The method itself is consistent with the aim of the report and the description is sufficient that the reader can assess that it would produce data that could answer the RQs.					
						The methods follow closely to approaches described in the lab script and give additional detail that is appropriate, relevant, and necessary for the reader to interpret the results.		The methods of data collection or analysis include some methods that are not provided in the lab script, while still being a physically sensible approach to extracting quantities of interest.		Novel methods for measuring quantities not described in the lab script are given and are fully documented such the reader could easily reproduce them.	
5. **Justification** of methods used (the choice of apparatus & associated parameters & data analysis approaches)	The method used to collect data was not described.	No justification is provided for the methods used in the data collection, or the method was used because that is what the lab script said to do.		Some effort was given in justifying the methods used to collect and analyse data including **identification** of assumptions and limitations.		The choice and development of methods to collect and analyse data is justified by considering (correctly) the **effect** assumptions and limitations would have **on the measured data and results**.		The choice of method used to collect **and** analyse data is described including how it was calibrated/optimised and any assumptions that can be justified correctly based on theoretical considerations and/or apparatus limitations. At least one assumption/choice in the method is **tested** (experimentally or with a simulation) to identify any bias it might introduce into the conclusions.		+ All pertinent assumptions/choices in data collection **and** analysis have been tested quantitatively and a logically consistent argument is provided to justify how the effect of the assumption is factored into the interpretation of the data and the effect on the result.	

Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		1st			
Data											
		Partial	Complete	Partial	Complete	Partial	Complete	Partial		Complete	
6. Quality of data collected (plots/tables/ inline)	No plots or data tables.	Cannot determine the quality of the data. E.g., points are not clear, or captions do not properly label the data so it is not clear where it has come from or what is supposed to be represented.		Low-quality data: data that is under-resolved or clearly does not show what is being claimed to be measured. Some quantities are missing uncertainties.		Medium-quality data: some, but not all, data meets the criteria of high-quality data. Uncertainties are provided on all measured quantities.		High-quality data: data that is well-resolved given instrument/ methodological limitations. Plots include error bars where appropriate. Plots clearly indicate the trend/relationship that is being claimed in the main text. Key results can be determined from the figures and captions alone.			
Conclusions											
Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		1st			
		Partial	Complete	Partial	Complete	Partial	Complete	Partial		Complete	
7. Conclusions	There are no conclusions.	Conclusions do not answer the research question or aim.		Conclusions are **not** based on fully correct physics but do provide a quantified response to the RQs. OR Conclusions are based on correct physics but do **not** provide a **quantified** response to the RQs.		Conclusions are based on **valid physics** and link back and provide a quantified response to the research question.		+ Conclusions provide valuable insights to the reader into the physics of the system being studied.			
Descriptor	Unclassified	3rd		Lower 2nd		Upper 2nd		Lower 1st		Upper 1st	
		Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete	Partial	Complete
8. Evaluation of conclusions through uncertainty analysis	There is no uncertainty analysis or evaluation of the conclusions.	A qualitative evaluation of the conclusions is provided with little supporting evidence. Uncertainties on the final result have not been considered.		The uncertainties in the final measured/calculated values are used in the evaluation of the conclusions. These uncertainties are not entirely consistent with the data and analysis presented.		Conclusions are justified based on the data and analysis presented. Implications of the uncertainty associated with the results on the validity of the conclusions are considered.		Conclusions are justified based on the data and analysis presented. The **size** of effects/differences and appropriate tests of the **statistical significance** of results are calculated and the implications of quantified systematic and random uncertainties arising from the methods chosen on the validity of the results is discussed.		+ Conclusions are consistent with **all** statements made and the limitations of the experiment. The answer to the RQ is articulated in terms of a hypothesis that has been tested and an appropriate method (relevant to the practice of the field of physics) of testing that hypothesis is used.	

Similar rubric as 2^{nd} year lab:

- You will receive a grade for each category
- Not a numerical mark

Feedback:

- Statements associated with the grade (rubric)
- Annotation on the report
- Statement on area to improve
- Statement on good practise

Written independently?

- Present the exact same figures
- Quote the same numerical results
- High degree of similarity in references*
- **ALL** text must be written independently
  - main text
  - figure captions
  - *placement of references

- Submission through TurnItIn on Blackboard, which checks
  - database of published work
  - database of websites
  - database of student work
  - previously submitted reports
  - current submitted reports

https://www.imperial.ac.uk/physics/students/current-students/undergraduates/degree-programmes-modules-and-timetables/plagiarism/

Record keeping – the lab notebook

- Real experiments can take months or years to complete. You will **never remember all the critical details**
    - 4th year projects, MSc project, PhD

- Research projects are often carried out in large teams that roll-over. Needs to be a **clear and permanent record** of work done, experimental details, operating procedures, etc., preserved

- In industrial and academic research, considerable prestige and money can be involved in who did what and when. For this reason lab books are **legal documents of record**

- The lab book is an invaluable tool when **discussing research** (i.e. with demonstrators!) that should contain all salient details

Record keeping – the lab notebook

Your lab book is a record of what you see, measure and do in the lab. It should be:

- **Complete**: everything goes in – including the "mistakes"
- **Clear**: it need not be pretty, but it must be clear / legible
- **True/accurate**: do not embellish it, do not record things the way you think they should be, record them as they are!
- **Contemporaneous**: It should be a real time record, written as it happens, not written up neatly later. Don't leave sections blank to fill in later!

Imperial College London

Physical or Electronic Lab Notebook

- A **bound or stapled notebook**, preferably with a durable cover
  - my firm recommendation
  - no refill or spiral pads, ring-binders, scraps of paper…

- An **appropriate** note taking application
  - must time-stamp entries and edits
  - must allow multiple entry types (text, pictures, freehand, etc)
  - must convert to appropriate document format (e.g. pdf)
    - OneNote
    - Labstep
  - College webpage: https://www.imperial.ac.uk/research-and-innovation/support-for-staff/scholarly-communication/research-data-management/organising-and-describing-data/eln/

Imperial College London

What can you expect in the 3rd year lab?

- Experiments have scripts and usually begin with a short introductory talk to orientate you and provide key guidance – not intended to give step-by-step advice so expect to think for yourself.
- You may need to find additional things (cables, connectors, samples, data sheets, etc) away from the setup to make things work properly.
- Likely to take time to make things work well enough to take “good” data. **95% Rule ... Most of what you do does not result in quality data!**
- You need apply what you have learnt and go **beyond the script’s core experiment** to get a very good mark.

Imperial College London

Demonstrator Interactions evolve, *it's a feature not a bug*.

	Experimental Question	Demonstrator Response	Theory Question	Demonstrator Response	Context
1st Year	How can I make single piece of kit X do job Y?	Watch how X change as I adjust this bit Y, now you try.	What does term X in equation from script mean?	Term X has units Y and describes well understood Z.	Focus on a single immediate issue, few carefully chosen instruments guaranteed to work. Obvious single value answers, detailed guidance from complete script, hands on demonstrator assistance.
2nd Year	How can I make my 2-3 carefully selected instruments do X?	You should try single thing X and see how Y changes.	What does this equation X from the script mean?	It describes physical process X, you will use it in obvious way to find Z.	Well defined system of a few elements, carefully chosen to work together, often pre configured to produce useful results, minor tweaks yield well defined data sets with obvious theoretical underpinning. General guidance from script with substantial detail, some hands on demonstrator support. Experiment will work if you are careful.
3rd Year	How can I make my collection of multiple unfamiliar instruments do X, Y and Z?	Tell me your strategy for solving this problem in multiple ways, watch out for tricky thing X when you do. Show me trial data.	This equation seems incomplete!	Well spotted, you should consider special case A and perhaps non-ideal behaviour B that the equation missed out.	More complex systems must be put together in correct way, optimised, calibrated, before giving useful results. Decision trees appear, methods X, Y and Z will work, you select and test to find best approach. Usually poor results at start, must repeat and refine technique. Script gives guidance, intentionally leaves out some key detail that exists, requires hunting in books/spec sheets etc. Demonstrators more challenging of your approach, asking you to explain and justify what you are doing. Your experiment should work, but typically takes several attempts to get right. Expect initial data to be nothing more than a signpost for how to do better.

Imperial College London

What should you expect from 3rd year demonstrators?

- **More challenge!** You should be learning to work independently. If you encounter a problem, try to solve it yourself first.
- If you are ‘lost’, ask a demonstrator - don’t expect easy answers. They are trained to guide you to find a solution yourself so may well reply with a well-targeted question. **It’s a feature, not a bug.**
- Do share your progress with your demonstrator - a good lab book with up to data plots really helps here.
- When the demonstrator asks you how you are getting on, tell them where you are, what you have found, discuss how you will analyse your findings, the underlying physics, mathematics, etc.

Imperial College London

Questions ?